Noise Removal versus Signal Loss in Network Filtering of Daily GNSS Time Series: Internal Working Report, Pacific Northwest, Validation Years

H. Oğuz Bolat

[a] *Department of Geomatics Engineering, Yıldız Technical University, Istanbul, Türkiye*

Abstract

This report describes a signal-loss audit of network filters for daily GNSS position time series on the Pacific Northwest network and reports its results on the validation years. Daily IGS20 solutions of the Nevada Geodetic Laboratory for 472 stations between 42.5° and 49.5°N are reduced to residuals that keep each station's seasonal cycle and its earthquake offsets. Thirty-five filter configurations are audited: target-inclusive filters (network stack, sub-regional probabilistic PCA, robust weighted median), distance-excluded filters (the mean of every station beyond a radius R), leave-one-out filters over nearest neighbours, and learned filters, from a learned kernel to space–time attention networks trained with and without planted transients, each learned filter with three training seeds. Each is scored on the noise it removes, on its leave-one-out error on fixed masks, on the fraction it keeps of 711 planted Gaussian transients 10–100 km wide and on the spurious signal it creates outside them, and on three real signals without ground truth: the seasonal cycle, one coseismic offset and 10 slow-slip episodes. On the validation years (2020–2021) at $R = 0$, the far stack removes 26.8% of the short-period scatter, has a mask error of 0.302 and keeps 0.985 and 0.774 of 25- and 100-km transients. The learned kernel over 256 neighbours has the lowest mask error, 0.292, and keeps 0.78 and 0.30. The attention networks trained with planted transients reach 0.297–0.304 and keep 0.92–0.95 and 0.61–0.66; trained without them, 0.296–0.303, 0.81–0.87 and 0.37–0.47. Across seeds, the mask error of a learned filter varies by at most 0.004 and its fraction kept of 100-km transients by up to 0.15. Every filter keeps 0.06–0.21

Email address: oguz.bolat@std.yildiz.edu.tr (H. Oğuz Bolat)

of the seasonal cycle, including the far stack at 400 km, which keeps every planted transient; about nine tenths of that cycle is shared by the whole network. Of the slow-slip displacement estimated against stations far from the tremor band, the far stack keeps 0.84, the filters over nearest neighbours 0.25–0.51 and the learned filters 0.36–0.55. Thirty-three of the 35 filters report the one real offset as larger than observed. Under the success criterion fixed on 12 November 2026 (noise removed higher than the far stack's with a bootstrap interval excluding zero, and 100-km retention and spurious signal within 0.05 of it), only the network mean that includes the target passes, by 0.002 in noise removed; no learned filter passes. The test years have not been scored.

Keywords: GNSS time series, common-mode error, spatial filtering, transient deformation, signal preservation, Cascadia

---

Status of this document. This is an internal working report, kept as a reference while the study is developed. It replaces the report on the Californian network (14 October to 5 November 2026), whose numbers were produced with the seasonal cycle removed from the series and were declared superseded on 5 November 2026; that version remains in the repository history (commit `c4b0084`). This version describes the audit on the Pacific Northwest network as it was run on 5–7 November 2026, and reports the validation years (2020–2021) only. The test years (2022 onward) have not been scored or examined and no choice was made on them, although the screening steps of Section 3.2 read their days; the method is not frozen: every number here is development evidence, and settings may still change. The text was drafted on 7 November 2026 by Claude Opus 5.5 (Anthropic) through Claude Code, at the author's request, from the repository and the results files on the computation machine. Every number was produced by `notes/internal/work/summary.py`, whose output is `notes/internal/work/summary.txt`. Sections 1–6 state what was done and what was measured. Section 7 lists questions that the numbers raise; it does not interpret the results, which is left to the author. The figures of the Californian report are not reproduced; they have not been regenerated for this network. Revised on 12 November 2026 by Claude Fable 5.2 (Anthropic) through Claude Code, at the author's request: the slow-slip episode measure was redefined and scored again for every filter, the far stack was run at 10 km, bootstrap intervals, the author's success criterion and the neighbour rule were added (Sections 4.2, 4.4, 6.7, 6.9 and 6.10), and the open questions were updated. All other numbers are those of 7 November. Training of seeds 1 and 2 at every radius and the signal tests at every radius were started on 12 November and are not in this version.

1. Introduction

GNSS stations record crustal deformation together with noise. Part of the noise is common to many stations of a network (common-mode error), and arises from errors in satellite orbits and clocks, in the realisation of the reference frame and from large-scale environmental loading (Wdowinski et al., 1997; Dong et al., 2006; Kreemer and Blewitt, 2021). Common-mode error has been estimated by stacking (Wdowinski et al., 1997), by principal or independent component analysis (Dong et al., 2006; Gruszczynski et al., 2018; Zhou et al., 2022), and by weighted means and medians of neighbouring stations (Tian and Shen, 2016; Kreemer and Blewitt, 2021). The evaluation reported for these filters is most often the reduction in the root-mean-square

(RMS) scatter of the filtered series (Tian and Shen, 2016; Kreemer and Blewitt, 2021; Zhou et al., 2022). Kreemer and Blewitt (2021) note that the main reason their filter reduces scatter more than traditional stacking is that it absorbs local common-mode transients.

Learned filters have been evaluated in other ways. Bachelot et al. (2025) predict each station from its eight nearest stations beyond 400 km, a distance chosen empirically in their study area as the shortest at which slow-slip offsets were no longer reduced by denoising, and they assess signal preservation on 280 slow-slip offsets measured on raw and denoised series. Costantino et al. (2024) add synthetic slow-slip events to noise generated from real GNSS series and report the error of the recovered displacement against the signal-to-noise ratio. Mastella et al. (2025) train a single-station method on synthetic series and evaluate the noise removed from real series against the residual of a trajectory-model fit (Bedford and Bevis, 2018). Tian and Shen (2016) estimated the common mode from stations beyond increasing distance thresholds and used the threshold at which the RMS changed most sharply as the approximate extent of a regional transient. Gualandi et al. (2016) evaluated source separation on simulated networks with known sources in synthetic noise. In Cascadia, Costantino et al. (2026) report that the scaling, b-value and stress-drop distribution of slow-slip events change with the detection threshold applied to slip inverted from denoised data, and the debate over a reported precursory acceleration of slip (Bletery and Nocquet, 2023, 2025) turned on whether a common-mode correction removes only noise.

The study described here measures signal loss directly. Transients of known extent, amplitude and duration are planted in the real residuals of the Pacific Northwest network; each filter is run on the residuals with and without them; and the difference between the two cleaned series gives the fraction of each transient that survives. The same runs give the noise each filter removes and the spurious signal it creates where nothing was planted. Each filter predicts a station from N other stations, excluding every station within a radius R of it; both N (from 16 stations to the whole network) and R (from 0 to 600 km) are experimental variables. The network lies above the Cascadia subduction zone, whose episodic slow slip moves the surface by a few millimetres over weeks along hundreds of kilometres of the margin (Michel et al., 2019), the kind of wide, coherent signal that averaging filters are expected to absorb. Section 2 reviews the literature, Section 3 describes the data and their processing, Section 4 the audit, Section 5 the filters (Table 3), Section 6 the validation-year results, and Section 7 lists

open questions.

2. Literature review

This section is condensed from the research proposal. Two works cited there are left out: a blog post, and a study the author has not read in full.

2.1. Classical common-mode filtering

Network filtering of GNSS data began with the Southern California Integrated GPS Network, where the common-mode error was estimated as the weighted mean of the residuals at the other stations and subtracted from each series (Wdowinski et al., 1997). Principal component analysis and the Karhunen–Loève expansion extended this to common modes that vary in space (Dong et al., 2006). Later work made principal component filtering applicable to networks whose stations start and stop at different times: a modified analysis that reproduces each series from its principal components without filling gaps (Shen et al., 2014), probabilistic principal component analysis that treats missing days as latent values (Gruszczynski et al., 2018), and a variational Bayesian version that also selects the number of components (Li et al., 2020). Because the common mode varies in space, large networks have been divided into sub-regions; on 208 Chinese stations, adaptive sub-regional filtering reduced residual scatter by 20–29%, against 15–18% for a single global analysis (Liu et al., 2026). Independent component analysis offers an alternative decomposition (Gualandi et al., 2016; Zhou et al., 2022).

Tian and Shen (2016) weighted each neighbouring station by its distance and by the correlation of its residuals with the target, chose the weights by a grid search that minimises the residual scatter after filtering, and outperformed principal component analysis by more than 13% in scatter reduction on about 1,600 stations in western North America. They also stepped through minimum-distance thresholds to separate long-range from regional common modes and to estimate the extent of regional transients. The common-mode component imaging of Kreemer and Blewitt (2021) takes a correlation-weighted median over nearby stations, including the target itself when it qualifies, and reduced residual scatter by 39–50% on about 2,500 stations in western Europe. Across this family, performance is reported as scatter reduction, reduced inter-station correlation, velocity uncertainty or the recovery of a simulated common mode.

2.2. Distance-excluded and learned filters

The graph neural network of Bachelot et al. (2025) represents stations as nodes and connects each to its eight nearest stations beyond 400 km. During training, 30% of the nodes are masked and the error is computed on the masked nodes only, so the network learns to predict a station from distant stations, and the prediction is subtracted as noise. They compared it with three stacks, including a stack of all stations beyond 400 km, which performed comparably and better on the network-average metric. The 400 km was chosen as the shortest distance at which 280 slow-slip offsets from 15 Cascadia events were no longer reduced by denoising.

Costantino et al. (2024) train a network of graph recurrent layers and transformers to extract slow slip from synthetic examples, slip on the subduction interface added to noise synthesised from real Cascadia series by principal component decomposition and Fourier phase randomisation; validated by synthetic recovery and agreement with tremor, it was later used to invert fifteen years of denoised data (Costantino et al., 2026). Mastella et al. (2025) model high-frequency scatter at single stations, with noise defined as the residual of a trajectory decomposition (Bedford and Bevis, 2018) and synthetic training noise from an adversarial network. Teutschmann et al. (2026) pretrained a self-supervised foundation model on 17,651 NGL stations by masked latent prediction, station by station, and fine-tuned it for displacement forecasting and the localisation of seismic offsets.

Predicting a withheld sensor from its neighbours is established elsewhere: tilt and compliance noise is removed from ocean-bottom seismometers by predicting the vertical channel from other channels (Crawford and Webb, 2000; Janiszewski et al., 2019), distributed acoustic sensing records are denoised by predicting each channel from its neighbours (Batson and Royer, 2019; van den Ende et al., 2023), and graph networks for spatiotemporal kriging are trained by reconstructing withheld sensors (Wu et al., 2021; Cini et al., 2022; Marisca et al., 2022). Excluding nearby observations so that spatial autocorrelation does not inflate apparent skill is the basis of buffered spatial cross-validation (Le Rest et al., 2014).

2.3. How signal preservation has been assessed

Three approaches recur (Table 1). Observed transients can be compared before and after filtering (Bachelot et al., 2025); this uses real signals but has no ground truth. Synthetic signals can be added to synthetic noise (Costantino et al., 2024; Mastella et al., 2025); this has ground truth but

depends on how realistic the noise is, and has been analysed against signal-to-noise ratio rather than spatial scale. Independent data such as tremor can confirm that an extracted signal is real (Costantino et al., 2024), but not measure what was lost. Detectors of transients are commonly trained on synthetic signals: Costantino et al. (2023) injected 60,000 synthetic slow-slip events computed with elastic dislocations (Okada, 1992) into surrogate noise and recovered 35 of 40 catalogued events (Michel et al., 2019); other approaches include training on real labelled windows (Tanaka et al., 2025), autoencoders for InSAR time series (Rouet-Leduc et al., 2021), the network strain filter (Ohtani et al., 2010), Kalman smoothing with principal components (Ji and Herring, 2013) and sparse estimation (Riel et al., 2014).

2.4. Gaps

Four gaps motivate the study. No study has measured, with ground truth, how much of a transient a common-mode filter removes as a function of the transient's size relative to the filter's reach. The reach has been chosen empirically, as a fixed 400 km (Bachelot et al., 2025) or by scatter-based thresholds (Tian and Shen, 2016). Filters that use the target station, which include most classical methods, have not been compared with leave-one-out filters on signal loss. And learned filters have been compared with simple ones on noise reduction, not at equal signal kept.

3. Data and processing

Figure 1 shows the processing chain. Daily positions are reduced to residuals by removing each station's trajectory, screened, and stored in a single array (the cube). Every filter is applied to the cube, and the cleaned series are scored by the audit of Section 4. All processing is done by the `gnssdl` package (`gnssdl build` for the cube, `gnssdl bench` for filtering and scoring).

3.1. Data

The data are the daily IGS20 solutions of the Nevada Geodetic Laboratory (NGL) (Blewitt et al., 2018; Nevada Geodetic Laboratory, 2026). Each solution gives the east, north and up position of a station's antenna relative to a fixed reference point near the station, estimated from 24 hours of observations, with formal uncertainties of about 1 mm in the horizontal and

Table 1: How signal preservation has been assessed in common-mode filtering and GNSS denoising.

Study	Test of signal preservation	Ground truth	Dependence on spatial scale
Tian and Shen (2016)	Minimum residual scatter; scatter against distance threshold	None	Transient extent inferred, loss not measured
Kreemer and Blewitt (2021)	Scatter reduction, velocity repeatability	None	Absorption of local transients noted, not measured
PCA and ICA filters[a]	Simulated common mode, inter-station correlation, velocity uncertainty	Noise only	Not examined
Mastella et al. (2025)	Synthetic trajectories in synthetic noise	Synthetic	Single station
Costantino et al. (2024)	Synthetic slow slip in synthetic noise; tremor	Synthetic	Against signal-to-noise ratio
Bachelot et al. (2025)	280 observed slow-slip offsets before and after	None	One distance (400 km)
This study	Transients planted in real noise; seasonal cycle, real offsets and slow-slip episodes	Planted	Retention and spurious signal against width and exclusion radius

[a] Shen et al. (2014); Gruszczynski et al. (2018); Li et al. (2020); Zhou et al. (2022); Liu et al. (2026).

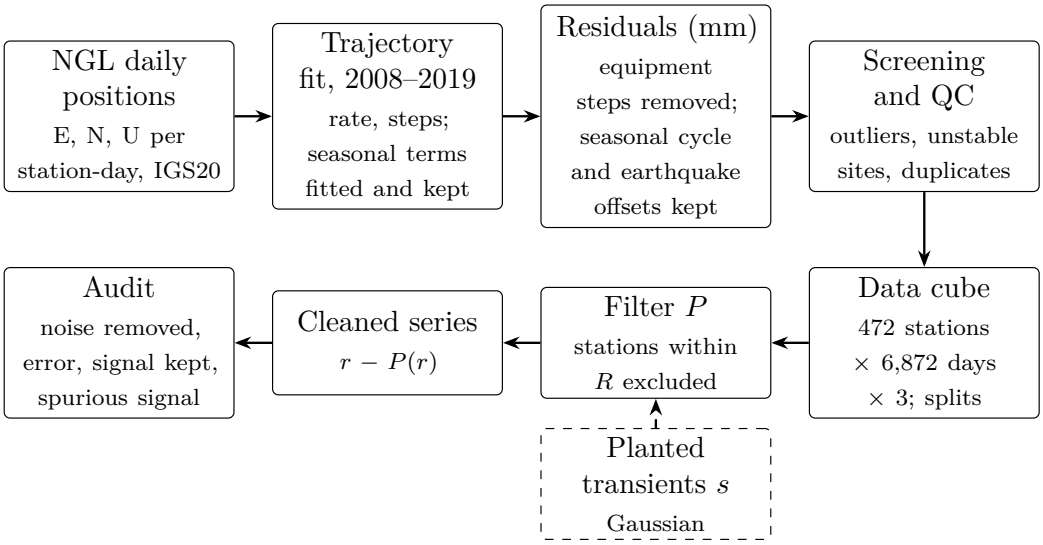


Figure 1: Processing chain from daily GNSS positions to the signal-loss audit.

3–4 mm in the vertical. NGL publishes new days with a latency of about ten days.

All NGL stations between 42.5° and 49.5°N and between 125.5° and 116.0°W with at least 1,095 daily solutions were selected on 5 November 2026: 540 stations of the Pacific Northwest Geodetic Array, the Network of the Americas and other networks. Their median distance to the nearest station is 6.2 km. The cube built on 5 November 2026 keeps 472 of them: 5 had no baseline, 16 were removed by the station quality control of Section 3.2 and 47 were co-located twins within 1 km of a station with more daily solutions. It covers 6,872 days, from 1 January 2008 to 24 October 2026, with 69% of the station-days available (77% in the validation years), and 432 stations have data in the validation years. A station has a median of 244 stations within the 300-km neighbour search at $R = 0$ and 132 at $R = 400$ km (Table 2).

3.2. Residuals

For each station i and component c, the trajectory model

$$x_{i,c}(t) = a + v\,(t - t_0) + \sum_{k=1}^{2} \left[p_k \sin(2\pi k t) + q_k \cos(2\pi k t)\right] + \sum_j b_j\, H(t - t_j) + r_{i,c}(t) \tag{1}$$

Table 2: The study network and the stations scored on the validation years.

Stations selected (box, $\geq$ 1,095 solutions)	540
Stations in the cube	472
Held-out stations (farthest-point sampling and short records)	77
Days	6,872 (2008-01-01 to 2026-10-24)
Training / validation / test days	4,383 / 731 / 1,758
Station-day coverage, all years / validation years	69% / 77%
Stations with data in the validation years	432
Stations left out of scoring by the scoring QC	8
Stations scored for noise removed	395
Stations scored for seasonal kept	372
Mask cells, `scatter` / `block` / `station`	22,374 / 18,576 / 37,501
Mask stations, `scatter` / `block` / `station`	333 / 339 / 72
Training station-days barred by the slow-slip exclusion	13.5% (251 sta.)
Median stations within 300 km at $R = 0$ / 400 km	244 / 132

is fitted by least squares on the training years (2008–2019), where a is a constant, v a linear rate about the mean epoch t_0 of the fitted data, p_k and q_k the amplitudes of the annual and semi-annual harmonics, b_j the amplitudes of Heaviside steps H at the epochs t_j of the NGL step catalogue, $r_{i,c}(t)$ the residual and t the time in years. A step is estimated only where at least 30 epochs lie on each side of it. A station with fewer than 365 daily solutions in the training years is fitted on the first two years of its record, or on its first 365 solutions if these reach further; such stations are always among the held-out ones (Section 3.3). The fitted trajectory is extrapolated to the later years.

The residual passed to the filters is the series minus the constant, the rate and the equipment steps. The seasonal terms are estimated so that an incomplete cycle does not bias the rate, and then added back, so the residual carries each station's seasonal cycle; what a filter does to it is measured directly (Section 4.2). Earthquake steps are likewise estimated and added back, so that coseismic offsets remain in the series. Equipment steps are removed everywhere: within the training years they are estimated in the fit, and after them by a local estimate, the median residual over the 30 days after the change minus that over the 30 days before, with at least 5 days on each side.

A value more than five robust standard deviations (1.4826 times the running median absolute deviation) from its 61-day running median in any component is treated as missing. For station screening only, a second fit is made from which the earthquake steps and the seasonal terms are also removed, so that real offsets and the seasonal cycle do not count as noise; a station is dropped if the robust scale of this fit exceeds 10 mm in the horizontal or 30 mm in the vertical, which removes non-tectonic site motion. The second fit changes no series.

Stations that develop problems after the training years are kept as inputs but not scored (scoring QC): those whose day-to-day scatter in the validation and test years exceeds three times their training-years scatter, and those with a jump of more than 50 mm in the horizontal or 100 mm in the vertical between consecutive 30-day medians with no catalogued step within 45 days. Eight stations are left out of scoring, all for unexplained jumps; the first jump falls in the validation years at two of them (CLHQ, January 2020; BLYN, August 2020) and in the test years at six (BCNN, MGRB, P688, IDFB, CPCO and SEPR, 2022–2025). The scoring QC reads each station's own series in the validation and test years and nothing else, and is the same for

every filter.

The residuals, their formal uncertainties and an availability mask are stored in the cube, with the neighbour lists of every station at each exclusion radius. Days are split into training (2008–2019), validation (2020–2021) and test years (2022 onward). The local equipment-step estimate, the outlier screening and the scoring QC use days from every year; their effect on a day that is kept is at most a constant shift or the removal of a spike.

3.3. Held-out stations and excluded intervals

The held-out stations are never used in training, and serve to test how well trained filters generalise to stations they have not seen. Ten percent of the stations are chosen by farthest-point sampling (Gonzalez, 1985) with seed 0, which starts from a random station and repeatedly adds the station farthest from all those already chosen; the stations with fewer than 365 training-year solutions are added, giving 77 held-out stations, 35 of them short-record stations. The trajectories of the short-record stations are fitted on the first part of their record, which can extend into the validation or test years.

The Cascadia slow-slip episodes of the training years are excluded from training, so that no filter is fitted or trained on them: the episodes of the GNSS catalogue of Michel et al. (2019), each over its along-strike extent from the paper's source data, bar 13.5% of the available training station-days, at 251 stations. Learned filters are in addition trained only outside ±30 days of catalogued M $\geq$ 6 earthquakes at a station.

3.4. Displacement rather than increments

Filters are applied to displacement rather than to daily increments. For example, 2 cm of postseismic displacement accumulated over a year is a daily increment of about 0.055 mm, much smaller than the day-to-day scatter of the increments. Increments are used only as an additional input to the M5 attention network (Section 5.4).

4. The signal-loss audit

4.1. Leave-one-out filtering and the exclusion radius

A filter P estimates the part of the residual of station i that is shared with other stations, and the cleaned series is $r - P(r)$. A leave-one-out filter never reads station i itself when estimating it, and never reads any station within R km of it. The target-inclusive filters of Section 5.1 read the target

on purpose, as published filters of their kind do; they are applied to the same cube and scored on noise removed and on the signal measures, but not on the masks. The exclusion radius R is an experimental variable. The far stack is run at $R = 0$, 10, 25, 50, 100, 200, 400 and 600 km; the nearest-neighbour filters (M1 to M4) and the learned filters are trained and scored on the masks at the cube's radii of 0, 10, 25, 50, 100, 200 and 400 km, the whole-network M6 variants up to 100 km. The nearest-neighbour filters are run through every signal test at every radius; the learned filters at $R = 0$ and 25 km with seed 0 and at $R = 0$ with seeds 1 and 2.

Two leak checks are run for every leave-one-out filter at every radius, on the first year of the validation period. Both compare predictions on two inputs that differ only in values the filter must not read. In the radius check, the target and every station within R of it are set to a poison value of 10^6 mm, for five randomly chosen targets. In the mask check, the cells hidden for the mask scores are filled once with missing values and once with the poison value, while remaining marked as unavailable. In both, the predictions must agree to within 0.001 mm. All 183 recorded checks passed, the 118 training runs of the overnight queue of 6–7 November 2026 among them. Because a filter that passes the radius check never reads the target, one prediction on the unhidden cube is a leave-one-out prediction for every station at once, and this is how all cleaned series are produced.

4.2. Measures

Filters are compared on two measures of noise and five of signal, all computed on the validation years. The planted transients are the primary signal test; the real signals are secondary checks without ground truth.

Noise removed is the reduction in short-period scatter at each scored station,

$$1 - \frac{\mathrm{rms}\Big(\mathrm{fast}(r - P(r))\Big)}{\mathrm{rms}\Big(\mathrm{fast}(r)\Big)}, \tag{2}$$

where the fast part of a series is the series minus its 61-day running median. It is averaged over the three components and summarised by its median over the 395 scored stations. Predicting zero gives 0. The measure covers the fast band only; common-mode error at periods longer than about two months earns no credit here.

The mask error is the leave-one-out error on fixed evaluation masks (seed 20260101) of the validation days: 10% of the available station-days of the

seen stations at random (`scatter`), one 30-day block per seen station and year (`block`), and every day of the held-out stations (`station`). At each hidden cell the error $r - P(r)$ is taken, its fast part is formed, its root mean square is normalised by each station's robust noise scale from the training years, and the result is averaged over the three components and summarised by its median over stations. Days within ± 30 days of an $\mathrm{M} \geq 6$ earthquake catalogued for a station are left out. Predicting zero gives 0.423 on `scatter` rather than 1, because separating the slow part also removes some of each station's own noise. The leave-one-out filters were tuned, and the learned filters early-stopped, on a related criterion, the same fast error over every available validation day of the seen, scored stations (the validation criterion of Table 11); the two are computed on different cells and are not equal. The error is lower when the prediction reproduces whatever the target shares with other stations, signal included, so tuning on it favours settings that absorb coherent signal; the signal measures below are independent checks on what the tuning selected.

Retention measures the fraction of a planted transient that survives filtering. A transient s is added to the residuals, and both the original and the modified residuals are filtered. The change in the cleaned series and the retention are

$$\Delta = \Big(r + s - P(r+s)\Big) - \Big(r - P(r)\Big), \qquad \rho = \frac{\langle \Delta,\, s \rangle}{\langle s,\, s \rangle}, \tag{3}$$

with the inner products over the horizontal components, the days of the transient and the scored stations within $2L$ of its centre, where L is its width. Pairing with the unmodified run removes the noise term; for a linear filter, $\rho = \langle s - P(s), s \rangle / \langle s, s \rangle$ exactly. $\rho = 1$ means that the transient is kept, $\rho = 0$ that it is removed, and $\rho > 1$ that the filter adds to it. Retention is summarised by its median over transients.

Spurious signal is the change created where nothing was planted. A Gaussian transient is set to zero beyond $3L$ from its centre (its footprint). At the scored stations outside the footprint whose cleaned series the transient can change (stations with a neighbour inside the footprint for the nearest-neighbour filters, stations whose far set includes one for the far stack, every station of the same region for C3, every station for C1 and for the learned filters that read the far stack or the whole network), the peak horizontal amplitude of Δ is taken as a fraction of the amplitude of the transient, the largest value over these stations is kept, and its median over transients is

reported, with the median number of such stations.

Seasonal kept fits, per scored station and component over the validation years, a constant, a rate, annual and semi-annual terms and Heaviside steps at the station's catalogued earthquakes to r and to $r - P(r)$ on the same days, and reports $\langle c_{\mathrm{kept}}, c_{\mathrm{obs}}\rangle / \langle c_{\mathrm{obs}}, c_{\mathrm{obs}}\rangle$ over the four harmonic coefficients c, per component (E, N, U), over the two horizontal components (H) and over all three. It is summarised by its median over the 372 stations scored. Unfiltered, the median annual amplitudes are 2.40, 3.75 and 6.14 mm in east, north and up, and the semi-annual 0.78, 0.91 and 0.86 mm.

Real offsets are every catalogued earthquake step in the scored period at a scored station with an observed horizontal offset of at least 3 mm and no other step within 15 days. The offset is the mean position over the 10 days after the earthquake minus that over the 10 days before (at least 5 available days on each side), and the fraction kept is $\langle \mathrm{kept}, \mathrm{observed}\rangle / \langle \mathrm{observed}, \mathrm{observed}\rangle$, with "kept" the same difference on the cleaned series. On the validation years there is one such offset: 4.89 mm at IDNR from the 31 March 2020 M6.5 central Idaho earthquake (catalogue identifier us70008jr5).

Slow-slip episodes are scored from a fixed list, with a measure revised on 12 November 2026. For each episode, a station's displacement d is its mean horizontal position over the 10 days after the episode's end minus the mean over the 10 days before its start (at least 5 available days in each window). The windows of the listed episodes lie 40 to 144 days apart, so d also contains the motion that the whole network shares over them. The far reference is the component-wise median of d over the scored stations more than 300 km east or west of the tremor band axis, at any latitude (at least 10 such stations). The slow-slip estimate is $s = d -$ reference, taken on the unfiltered residuals, at the scored stations within the episode's latitude range and within 150 km of the axis. The fraction kept of an episode is

$$\frac{\sum_i \langle \mathrm{kept}_i,\, s_i\rangle}{\sum_i \langle s_i,\, s_i\rangle}, \tag{4}$$

summed over these stations, with kept_i the displacement of the cleaned series, from which no reference is subtracted; this is the pooled form of Eq. (3). The noise floor of an episode is the median size of s over the far stations, where no slow slip is expected. An episode is counted if it has at least 10 scored band stations and their median size of s is at least twice the floor; the others are listed as weak and left out of the summary, which is the median over the counted episodes. A validation run may score an episode only if its

whole measure window lies on training or validation days; the episodes of the training years qualify because they are excluded from training (Section 3.3). The validation run scores 18 episodes of the GNSS catalogue of Michel et al. (2019); 15 have at least 10 scored stations and 10 of these are counted (932 station-episodes). The 31 tremor-timed episodes of 2022 onward lie in the test years and are not scored. Three properties of the measure follow from its construction. The unfiltered series does not score 1, because it still holds the reference motion, which projects on s by chance. A filter that removes exactly the reference scores 1. The estimate s is not ground truth: it holds each station's own noise and any difference in seasonal motion between band and far stations over the window, and the far stations all lie east of the band, so an east–west gradient in the shared motion counts as slip.

The first definition of this measure (5–7 November 2026) took the fraction kept of d itself, as the median over stations. A check on the training and validation days on 12 November showed that d was mostly network-wide motion: in most episodes the stations more than 300 km from the band had a displacement of similar size and direction to the band stations, and the plain network mean, which removes that motion, scored 0.18. The numbers of that definition are in the previous version of this report (commit `a018466`).

4.3. Planted transients

Gaussian transients have the form

$$s(d,t) = A \exp\left(-\frac{d^2}{2L^2}\right) g(t)\,\mathbf{u}, \tag{5}$$

where d is the distance of a station from the centre, A the amplitude, L the width, $\mathbf{u}$ a horizontal unit vector in a random direction and $g(t)$ the time profile: a raised-cosine rise over D days, a 60-day hold and a raised-cosine fall over D days, after which the transient is zero, so it leaves no permanent offset. s is zero beyond $3L$, and every station in the footprint moves in the same direction. The grid is $A \in \{2, 5, 10\}$ mm, $L \in \{10, 25, 50, 100\}$ km and $D \in \{10, 30, 90\}$ days. Each of the 36 combinations is planted at 20 centres, each at a randomly chosen scored station, giving 720 transients per filter and radius, of which 708–712 have at least one scored station within $2L$ (711 for the far stack at $R = 0$). The median number of scored stations within $2L$ is 3, 10, 38.5 and 125.5 for $L = 10$, 25, 50 and 100 km. Onsets fall in 300-day slots of the validation years, and the signal is added only on days with data. Many transients share one run of a filter: two share a run only if no station's

cleaned series can be changed by both. The library of fault-slip events of the Californian development is not used on this network.

4.4. Protocol

All design decisions, hyper-parameters and model choices are made on the validation years. The scoring code scores only the validation split unless the test split is requested explicitly, and no filter has been scored and no signal test run on the test years of this network; their days are read only by the screening steps of Section 3.2 (the local equipment-step estimate, the outlier screening and the scoring QC). The method will be frozen, with a tagged version of the software, before the test years are scored once with the settings chosen on validation; until then nothing here is final. Each learned filter was trained with three seeds at $R = 0$.

Uncertainty is estimated by bootstrap (1,000 resamples, 95% intervals). For noise removed and for the leave-one-out error, stations and 30-day blocks are resampled, the same resample for the two filters of a comparison; for retention and spurious signal the planted transients are resampled; for seasonal kept the stations. The sums that the noise bootstrap needs are saved per station and block when a filter is scored, and the intervals are computed from the saved results.

The author fixed the following success criterion on 12 November 2026, replacing that of 5 November, which used the leave-one-out error and applied to learned filters only. A filter is reported as better than the far stack at $R = 0$ only if its noise removed is higher with a bootstrap interval that excludes zero, its retention of 100-km planted transients is not lower than the far stack's by more than 0.05, and its spurious signal for them is not higher by more than 0.05. A filter with a higher noise removed that fails a signal condition is reported as a noise gain bought with signal. The rule applies to every filter, the target-inclusive ones included. The best filter and its settings are the passing filter with the highest noise removed; passing filters whose difference from it has an interval including zero are reported as indistinguishable. Seasonal kept, the real offset and the slow-slip episodes are reported and are not part of the rule.

A filter over the N nearest stations reads up to N of them, those beyond R and within 300 km of it. A combination of R and N is flagged when more than half the scored stations have fewer than N; a station with fewer than four neighbours beyond R is not scored at that radius, for any filter. On this

network no station is excluded at any radius, and $N = 256$ is flagged at 0, 10, 200 and 400 km (Section 6.10).

From 12 November 2026 the software refuses to score the test years unless a freeze record names a version tag and the source code is unchanged since that tag.

Until 12 November 2026 the identifiers of the planted transients were assigned in an order that depends on the exclusion radius of the filter, so the same identifier named different transients for different filters (the set of transients was the same). No result in this report pairs transients by identifier. The planted-transient results reported here were produced before that date; paired comparisons of retention between filters require them to be produced again.

5. Filters

Table 3 lists the filters. No leave-one-out filter reads the target's own values on any day. The "days used" column gives the days of the other stations that a prediction for day t reads. The software and filter definitions are those of the Californian development; the settings marked as tuned were chosen again on the validation years of this network.

5.1. Target-inclusive filters

C1 is the stack of Wdowinski et al. (1997): the common mode of every station on a day is the unweighted mean of all available stations that day, the station itself included. C3 is sub-regional probabilistic PCA (Tipping and Bishop, 1999), applied as in Gruszczynski et al. (2018) with regions as in Liu et al. (2026): stations are grouped into 1, 4 or 8 regions by k-means on their projected coordinates; in each region and component, a probabilistic PCA with 1, 2 or 3 components is fitted by expectation–maximisation on the quiet training residuals (Section 5.4), normalised per station, with missing days treated as latent; on any day the component scores are estimated from every available station of the region, the target included, and the common mode is the reconstruction. The nine settings are labelled c3-kkpp for k regions and p components. M2-self is the robust weighted median of M2 (Section 5.3) with the target's own value added to the median, as in common-mode component imaging (Tian and Shen, 2016; Kreemer and Blewitt, 2021); it reuses M2's tuned settings.

Table 3: The 35 filter configurations audited, named as in the results tables. Stations read: the other stations a prediction for the target uses; "nearest" means the nearest stations beyond R. Radii: exclusion radii at which results are reported. Days used: days of other stations read by a prediction for day t (31 d: a 31-day window used only to screen outliers; "all" and 64 d include past and future days).

Name	Description	Stations read	Radii (km)	Days used
Target-inclusive				
C1	Unweighted mean of the whole network	All + target	–	t
C3	Sub-regional probabilistic PCA, nine settings	Region + target	–	t
M2-self	Robust weighted median of the neighbours and the target	16 nearest + target	–	t (31 d)
Distance-excluded				
Far stack	Unweighted mean of every station beyond R	All beyond R	0, 25, 50, 100, 200, 400, 600	t
D1	Far stack at $R = 400$ km, computed separately	All beyond 400 km	400	t
Leave-one-out over nearest neighbours				
M1	Distance-weighted mean, $\ell = 50$ km	16 nearest	0–400	t
M1, 64	M1 over 64 neighbours, $\ell = 100$ km	64 nearest	0–400	t
M1, 256	M1 over 256 neighbours, $\ell = 100$ km	256 nearest	0–400	t
M2	Robust weighted median after outlier screening, $\ell = 100$ km, $k = 2$	16 nearest	0–400	t (31 d)
M3a	Ridge regression per station, $\lambda = 0.1$	16 nearest	0–400	t
M3b	Ridge regression pooled, with neighbour geometry, $\lambda = 0.001$	16 nearest	0–400	t
M4a, M4b	M3a and M3b per period band	16 nearest	0–400	All
Learned (three seeds at $R = 0$)				
M3k	Learned kernel from distance, azimuth and noise level	16 nearest	0–400	t
M3k, 256	M3k over 256 neighbours	256 nearest	0–400	t
M5, -aug	Space–time attention, correction to the far stack	32 of 64 nearest + far stack	0–400	64 d
m6rN, -aug	Attention over the $N = 16$, 64 or 256 nearest stations, correction to the far stack	N nearest + far stack	0–400	64 d
m6, -aug	Attention over the whole network	All beyond R	0–100	64 d
m6r, -aug	m6 with the correction to the far stack	All beyond R	0–100	64 d

5.2. Distance-excluded filters

The far stack is the unweighted mean of every available station at least R from the target, the target excluded, per day and component. It has no parameters. D1 is the far stack at $R = 400$ km, the distance of Bachelot et al. (2025), computed separately; its results agree with those of the far stack at 400 km. Bachelot et al. (2025) use the eight nearest stations beyond 400 km in a graph neural network and compare it with a stack of the same stations; D1 uses every station beyond 400 km.

5.3. Leave-one-out filters over nearest neighbours

These filters predict the target from its K nearest stations beyond R, using only day t (M1–M3) or the neighbours' whole series (M4). Their settings were chosen on the validation years of this network by the validation criterion.

M1 is the weighted mean $\hat{r}_i(t) = \sum_j w_{ij} a_j(t) r_j(t) / \sum_j w_{ij} a_j(t)$ over the 16 nearest stations, with $w_{ij} = \exp(-d_{ij}^2/2\ell^2)/\bar{\sigma}_j^2$, $a_j(t)$ the availability and $\bar{\sigma}_j$ the station's median formal uncertainty over the training years, per component. The length scale ℓ was chosen from 10 km to infinity (a plain mean): 50 km for 16 neighbours and 100 km for 64 and 256. At $R = 0$ the validation criterion is 0.328 for 16 neighbours, 0.318 for 64 and 0.315 for 256 at the chosen ℓ, and within 0.004 of these values for every $\ell \geq 50$ km.

M2 follows Kreemer and Blewitt (2021): a neighbour's value on day t is dropped if it differs from that neighbour's own 31-day running median by more than k times its robust scale; the rest are combined by a weighted median with M1's weights, interpolated as in Kreemer and Blewitt (2021). With fewer than three neighbours left it falls back to M1. The tuned values at $R = 0$ are $\ell = 100$ km and $k = 2$.

M3 is ridge regression on the three components of the 16 neighbours, an availability bit per neighbour and an intercept, in units of each station's noise scale. Missing neighbours are set to zero and the remaining values rescaled by K divided by the number available, so that gaps do not bias the fit. M3a fits one weight vector per station and component ($\lambda = 0.1$ in units of the trace of the Gram matrix per feature); the 77 held-out stations use M3b. M3b fits one weight vector per component for all stations, adding each neighbour slot's log distance and azimuth and their products with the slot's residuals ($\lambda = 0.001$).

M4 applies M3 separately to four period bands that add up exactly to each series (under 7, 7–60, 60–400 and over 400 days), obtained with differences

of zero-phase Gaussian smoothers, and sums the four predictions; λ is chosen per band. Before filtering, each series' gaps are filled by linear interpolation of its own days, and a neighbour missing on a day contributes zero with its availability bit off. The band filters read the neighbours' past and future days. M4a is per station and M4b pooled, as M3a and M3b.

5.4. Learned filters

The learned filters are trained on quiet residuals: the residuals minus the earthquake offsets estimated in the training-years fit, on training days outside the slow-slip exclusion and outside ± 30 days of catalogued M $\geq$ 6 earthquakes, at stations that are not held out. The loss is the squared error in units of each station's robust noise scale. Training stops early on the validation criterion: the attention networks are evaluated every 500 steps and stop after 3,000 steps without improvement, M3k after six epochs without improvement. Every learned filter was trained with seed 0 at each of its radii and with seeds 1 and 2 at $R = 0$ (Table 11). The architectures and training settings are those of the Californian development, carried over unchanged.

M3k keeps M1's form and learns its kernel: $w_{ijc} = \mathrm{softplus}(f_\theta(g_{ij}, q_j))_c$, where g_{ij} is the neighbour's log distance and the sine and cosine of its azimuth from the target, q_j its log robust noise scale per component and record completeness, and f_θ a multilayer perceptron with two hidden layers of 32 units. The weights depend on where a neighbour is and what kind of station it is, never on its values, so once trained M3k is a fixed weighted mean of day t. It is trained over 16 and over 256 neighbours.

M5 is a space–time attention network. A sample is a target, its 32 nearest stations beyond R, chosen for each 64-day window among the 64 nearest by availability, and one summary node, the far stack. Each node has 17 input channels per day: residual, log formal uncertainty, availability, a flag for cells hidden by a training mask, the daily increment $r(t) - r(t-1)$ (zero unless both days exist), the station's log noise scale and record completeness, and the day of year (sine and cosine), all in units of the station's noise scale. A convolution cuts each node's window into 4-day tokens; three blocks then alternate attention over a node's own tokens (time) and attention over the nodes at the same token (space), the latter biased by each node's log distance, azimuth and type relative to the target ($d = 64$, four heads). The output is a correction added to the far stack, so training starts from the far stack. The target is hidden for the whole window, so predictions never read it. Prediction

uses 64-day windows overlapping by half, combined with a triangular taper, so a day's prediction reads neighbours' days before and after it.

M6 gives the network a wider view than M5's 32 neighbours and one summary node. Its inputs are the residuals alone, per station and day in units of the station's noise scale, and an availability bit; geometry enters only as a learned attention bias for each pair of stations, from their log distance and azimuth, so no absolute coordinates are seen. A convolution cuts each station's 64-day window into 4-day tokens, and three blocks alternate attention over a station's own tokens (time) and attention over all stations of the sample at the same token (space), with $d = 64$ and four heads. In the whole-network versions (m6, m6r) a sample is every station: a random share of 5–30% of the stations is hidden as targets, together with every station within R of a target, and the targets are predicted from the rest. In the N-nearest versions (m6r16, m6r64, m6r256) a sample is one target and its N nearest stations beyond R. In the far-stack residual variants (the `r` in the name) the output is a correction added to the far stack and the target's own input slot carries its far-stack series, as in M5.

The -aug versions are trained with planted fields. In half of the training samples a synthetic field is added to the inputs of every node, and to the far-stack series as it would enter that mean, but not to the targets, so copying it into a prediction increases the loss. For M5-aug the field is one Gaussian blob or dipole centred within $R + 300$ km of the target, 5–150 km wide, of 1–15 times the station's noise scale, lasting 5–60 days with a raised-cosine rise and fall, in a random horizontal direction; the M6 -aug versions add one to three such fields. The generator uses its own seed and shapes and runs on training days only, so it is separate from the evaluation transients.

5.5. Implementation and use of AI tools

The `gnssdl` package, its tests (160 automated tests on 7 November 2026, including the leak checks), the design documents and the figure code were written by Claude Opus 5.5 (Anthropic) through Claude Code, under the author's direction, from 6 October 2026. The research questions, the design decisions and the choice of study area are the author's. The code is openly available at `https://github.com/hogib/gnssdl` under the MIT licence; per-model results, logs and checkpoints are kept in `data/runs/<model>/` on the computation machine (24 cores, one 32 GB GPU). The learned filters were trained and all signal tests run there in a queue of 300 jobs from 00:26 on 6 November to 02:22 on 7 November 2026, all of which ended without error.

This report was drafted by the same tool from the repository and its results files.

6. Results

All results are from the validation years (2020–2021). Tables 4 and 5 give every measure for every configuration at $R = 0$ (D1 at 400 km), with the unfiltered residuals (m0) as a reference: Table 4 the noise measures and planted transients, Table 5 the real signals. For the learned filters, the tables give the mean over seeds 0, 1 and 2; Section 6.3 gives the spread.

6.1. Noise removed and mask error at $R = 0$

Unfiltered, the mask error is 0.423 on `scatter`, 0.431 on `block` and 0.470 on `station`. The far stack lowers it to 0.302, 0.308 and 0.347 and removes 26.8% of the fast scatter. C1, which also reads the target, removes 27.0%, and its signal measures differ from the far stack's by at most 0.004. The lowest mask error is that of M3k over 256 neighbours, 0.292 (seed mean), with the most noise removed of any leave-one-out filter, 28.4%. M1 over 256 neighbours and the attention networks without planted transients reach 0.296–0.298, except m6 (0.303); the -aug versions 0.297–0.304. M1, M2, M3 and M4 over 16 neighbours have errors of 0.314–0.323, higher than the far stack's, and remove 23.1–25.6%. Of the target-inclusive filters, C3 with 8 regions and 3 components removes the most, 29.5%, and M2-self 29.1%. Per component, the far stack's `scatter` error is 0.306, 0.231 and 0.369 in east, north and up, and that of M3k over 256 neighbours (seed 0) 0.294, 0.229 and 0.354.

6.2. Planted transients at $R = 0$

The far stack keeps 0.998, 0.985, 0.925 and 0.774 of transients 10, 25, 50 and 100 km wide, and C1 the same within 0.003. Its largest spurious change has a median of 0.028 of the planted amplitude, spread over a median of 423 stations. The filters over the 16 nearest stations keep 0.47–0.65 of 25-km and 0.03–0.21 of 100-km transients, M1 and M2 the least. More neighbours keep more: M1 keeps 0.025, 0.132 and 0.260 of 100-km transients over 16, 64 and 256 neighbours, and M3k 0.034 and 0.304 over 16 and 256. C3 keeps less as the number of regions grows: 0.73–0.76 of 100-km transients with one region, 0.43–0.48 with four and 0.21–0.22 with eight.

Table 4: Noise measures and planted transients for every filter configuration, $R = 0$ (D1: $R = 400$ km), validation years (2020–2021). Error: mask error on `scatter` (lower is better; target-inclusive filters are not scored on the masks). Noise: noise removed, Eq. (2). Kept: median retention by width L. Spurious: median over all transients of the largest spurious change, as a fraction of the planted amplitude; Sta.: median number of stations that can carry it. Learned filters: mean over three seeds.

			Kept, L (km)					
Filter	Error	Noise	10	25	50	100	Spurious	Sta.
m0 (unfiltered)	0.423	0.000	1.000	1.000	1.000	1.000	0.000	–
Target-inclusive								
C1	–	0.270	0.995	0.982	0.923	0.772	0.028	423
c3-k1p1	–	0.259	0.995	0.982	0.925	0.763	0.050	423
c3-k1p2	–	0.263	0.993	0.978	0.923	0.756	0.073	423
c3-k1p3	–	0.263	0.992	0.977	0.916	0.732	0.107	423
c3-k4p1	–	0.276	0.981	0.932	0.787	0.476	0.114	148
c3-k4p2	–	0.278	0.971	0.917	0.775	0.453	0.162	148
c3-k4p3	–	0.282	0.960	0.906	0.760	0.427	0.214	148
c3-k8p1	–	0.279	0.962	0.882	0.641	0.220	0.154	84
c3-k8p2	–	0.289	0.945	0.859	0.621	0.216	0.187	84
c3-k8p3	–	0.295	0.928	0.831	0.597	0.206	0.208	84
M2-self	–	0.291	0.899	0.532	0.147	0.026	0.200	36
Distance-excluded								
Far stack	0.302	0.268	0.998	0.985	0.925	0.774	0.028	423
D1 (400 km)	0.316	0.242	1.000	1.000	1.000	0.999	0.095	212
Leave-one-out over nearest neighbours								
M1	0.320	0.251	0.877	0.471	0.130	0.025	0.088	36
M1, 64	0.302	0.268	0.969	0.816	0.445	0.132	0.113	107
M1, 256	0.297	0.276	0.980	0.878	0.609	0.260	0.110	223
M2	0.319	0.231	0.952	0.585	0.166	0.030	0.218	36
M3a	0.323	0.240	0.929	0.635	0.320	0.204	0.105	36
M3b	0.318	0.239	0.925	0.602	0.217	0.086	0.120	36
M4a	0.318	0.256	0.947	0.652	0.327	0.205	0.109	36
M4b	0.314	0.245	0.931	0.608	0.248	0.110	0.119	36
Learned, mean of three seeds								
M3k	0.313	0.258	0.881	0.528	0.155	0.034	0.105	36
M3k, 256	0.292	0.284	0.927	0.777	0.567	0.304	0.108	223
M5	0.297	0.271	0.957	0.832	0.617	0.427	0.110	423
M5-aug	0.300	0.268	0.989	0.933	0.813	0.661	0.095	423
m6r16	0.297	0.274	0.959	0.807	0.590	0.435	0.092	423
m6r16-aug	0.298	0.270	0.988	0.916	0.779	0.632	0.101	423
m6r64	0.296	0.275	0.960	0.826	0.605	0.371	0.121	423
m6r64-aug	0.297	0.272	0.991	0.933	0.805	0.619	0.101	423
m6r256	0.296	0.273	0.957	0.846	0.677	0.428	0.116	423
m6r256-aug	0.297	0.271	0.989	0.942	0.825	0.622	0.099	423
m6	0.303	0.258	0.956	0.830	0.632	0.382	0.132	423
m6-aug	0.304	0.257	0.990	0.930	0.789	0.606	0.108	423
m6r	0.298	0.270	0.971	0.868	0.713	0.474	0.130	423
m6r-aug	0.300	0.266	0.997	0.954	0.840	0.661	0.153	423

Table 5: Real signals for every filter configuration, $R = 0$ (D1: $R = 400$ km), validation years. Seasonal kept: median over 372 stations, horizontal (H), vertical (U) and all three components. Offset: fraction kept of the one real offset (IDNR, M6.5, 31 March 2020; observed 4.89 mm). Episodes: median over the 10 counted slow-slip episodes of the pooled fraction kept, Eq. (4); the unfiltered series does not score 1 on this measure (Section 4.2). Learned filters: mean over three seeds.

	Seasonal kept				
Filter	H	U	All	Offset	Episodes
m0 (unfiltered)	1.000	1.000	1.000	1.000	0.888
Target-inclusive					
C1	0.100	0.103	0.125	1.457	0.833
c3-k1p1	0.066	0.101	0.124	1.419	0.786
c3-k1p2	0.043	0.102	0.101	1.475	0.720
c3-k1p3	0.043	0.104	0.093	1.471	0.594
c3-k4p1	0.065	0.102	0.102	1.371	0.619
c3-k4p2	0.041	0.102	0.089	1.350	0.485
c3-k4p3	0.040	0.090	0.077	0.812	0.394
c3-k8p1	0.067	0.111	0.109	1.310	0.494
c3-k8p2	0.040	0.104	0.090	1.271	0.388
c3-k8p3	0.037	0.094	0.083	0.404	0.331
M2-self	0.063	0.070	0.074	1.289	0.355
Distance-excluded					
Far stack	0.100	0.103	0.125	1.461	0.836
D1 (400 km)	0.097	0.170	0.169	1.469	1.006
Leave-one-out over nearest neighbours					
M1	0.080	0.081	0.095	1.259	0.357
M1, 64	0.086	0.074	0.091	1.324	0.437
M1, 256	0.090	0.082	0.095	1.324	0.509
M2	0.071	0.077	0.083	1.309	0.379
M3a	0.183	0.221	0.210	1.284	0.287
M3b	0.133	0.208	0.191	1.323	0.416
M4a	0.176	0.202	0.187	1.284	0.250
M4b	0.135	0.214	0.189	1.323	0.434
Learned, mean of three seeds					
M3k	0.089	0.118	0.117	1.308	0.361
M3k, 256	0.102	0.124	0.132	1.343	0.512
M5	0.062	0.049	0.062	1.429	0.371
M5-aug	0.082	0.069	0.079	1.454	0.483
m6r16	0.074	0.123	0.118	1.422	0.478
m6r16-aug	0.095	0.091	0.105	1.431	0.551
m6r64	0.079	0.087	0.096	1.462	0.439
m6r64-aug	0.104	0.097	0.106	1.477	0.510
m6r256	0.071	0.106	0.103	1.468	0.460
m6r256-aug	0.093	0.089	0.105	1.477	0.508
m6	0.081	0.124	0.114	1.481	0.429
m6-aug	0.088	0.104	0.104	1.464	0.504
m6r	0.053	0.109	0.095	1.461	0.457
m6r-aug	0.056	0.100	0.094	1.436	0.476

Table 6: Median retention of planted transients by amplitude and by duration, all widths, $R = 0$, seed 0, validation years.

	Amplitude (mm)			Duration (days)		
Filter	2	5	10	10	30	90
Far stack	0.96	0.97	0.96	0.96	0.97	0.96
C1	0.96	0.96	0.96	0.96	0.96	0.96
M1	0.23	0.25	0.24	0.24	0.26	0.23
M1, 256	0.73	0.78	0.77	0.77	0.78	0.74
M2	0.29	0.31	0.35	0.32	0.33	0.33
M3a	0.42	0.41	0.44	0.40	0.44	0.43
M3k, 256	0.70	0.72	0.70	0.69	0.73	0.70
M5	0.63	0.72	0.78	0.71	0.74	0.70
M5-aug	0.81	0.87	0.97	0.90	0.89	0.90
m6r16	0.62	0.70	0.75	0.71	0.71	0.66
m6r16-aug	0.69	0.85	0.96	0.89	0.88	0.88
m6r64	0.62	0.70	0.77	0.70	0.72	0.70
m6r64-aug	0.75	0.86	0.96	0.89	0.89	0.87
m6r256	0.67	0.74	0.80	0.75	0.76	0.73
m6r256-aug	0.77	0.91	0.99	0.93	0.93	0.94
m6	0.62	0.71	0.78	0.73	0.70	0.71
m6-aug	0.73	0.85	0.94	0.87	0.88	0.86
m6r	0.72	0.79	0.84	0.80	0.81	0.77
m6r-aug	0.81	0.89	0.94	0.89	0.91	0.89

The attention networks trained without planted transients keep 0.81–0.87 of 25-km and 0.37–0.47 of 100-km transients; those trained with them 0.92–0.95 and 0.61–0.66. Their largest spurious changes have medians of 0.09–0.15 of the planted amplitude, against 0.028 for the far stack, and every learned filter that reads the far stack or the whole network can change a median of 423 stations. No learned filter keeps as much of any width as the far stack.

For the linear filters, retention does not depend on the amplitude of the transient: the far stack keeps 0.96, 0.97 and 0.96 of 2-, 5- and 10-mm transients over all widths, and M1 0.23, 0.25 and 0.24. For the attention networks it does (Table 6): M5 keeps 0.63, 0.72 and 0.78, M5-aug 0.81, 0.87 and 0.97, and m6r256-aug 0.77, 0.91 and 0.99 (seed 0). Over all widths, retention differs by at most 0.05 between 10-, 30- and 90-day transients for every filter in Table 6. At $L = 25$ km, the filters over 16 neighbours keep more of 30-day transients than of 10- or 90-day ones (M1: 0.42, 0.55 and 0.44; M2: 0.50, 0.72 and 0.58).

6.3. Spread across training seeds

Table 7 gives the mean and the spread (largest minus smallest value) over seeds 0, 1 and 2 at $R = 0$. The spread of the mask error is at most 0.004 (median 0.002) on every mask, while the seed means of the learned filters span 0.292–0.313 on `scatter`. Noise removed varies by at most 0.005. The signal measures vary more: overall retention by up to 0.072, retention of 100-km transients by up to 0.149 (m6r256-aug: 0.699, 0.550 and 0.617), the largest spurious change by up to 0.072, seasonal kept by up to 0.094 (m6-aug: 0.050, 0.119 and 0.144) and episodes kept by up to 0.097 (M5). The difference between each attention network trained with planted transients and the same network trained without them, 0.11–0.17 in overall retention and 0.19–0.25 at 100 km (seed means), is larger than the seed spread of either. Among the attention networks trained without planted transients, the seed means of 100-km retention span 0.37–0.47, and among those trained with them 0.61–0.66, against seed spreads of up to 0.07 and 0.15.

6.4. Exclusion radius

Table 8 gives the mask error, noise removed, retention of 25- and 100-km transients and the largest spurious change of the far stack and the nearest-neighbour filters at every radius. As R grows, the filters keep more planted signal, remove less noise and create larger spurious changes, with the exceptions noted below. The far stack keeps 0.774 of 100-km transients at $R = 0$, 0.832 at 100 km and 0.999 at 400 km, while its mask error rises from 0.302 to 0.304 and 0.316 and its noise removed falls from 26.8% to 26.4% and 24.2%; at 600 km the error is 0.375. The filters over 16 neighbours keep 0.93–0.95 of 25-km transients at $R = 50$ km and 0.68–0.80 of 100-km transients at 200 km. M1 over 256 neighbours keeps less of 100-km transients at 50 km (0.315) than at 10 and 25 km (0.446 and 0.463), and its largest spurious change stays between 0.10 and 0.16 at every radius. The mask error of the learned filters at each radius (seed 0) is in Table 9; their signal tests were run at $R = 0$ and 25 km only. From 0 to 25 km, the learned filters keep 0.01–0.19 more of all transients and change their mask error by -0.001 to $+0.004$.

6.5. Seasonal kept

Every filter removes most of the seasonal cycle (Table 5). At $R = 0$, seasonal kept over all components lies between 0.06 (M5) and 0.21 (M3a); the far stack and C1 keep 0.125. Over the stations, the far stack's value has a median of 0.125 and 10th and 90th percentiles of -0.11 and 0.38. The ridge

Table 7: Learned filters over three training seeds at $R = 0$, validation years: mean (top) and spread, largest minus smallest (bottom). Scatter, block, station: mask errors. Noise: noise removed. Kept: median retention over all transients and at $L = 25$ and 100 km. Spur.: largest spurious change. Seas.: seasonal kept, all components. Epis.: slow-slip episodes kept.

Filter	Scatter	Block	Station	Noise	Kept	L 25	L 100	Spur.	Seas.	Epis.
Mean over seeds										
M3k	0.313	0.313	0.350	0.258	0.288	0.528	0.034	0.105	0.117	0.361
M3k, 256	0.292	0.299	0.339	0.284	0.669	0.777	0.304	0.108	0.132	0.512
M5	0.297	0.304	0.346	0.271	0.722	0.832	0.427	0.110	0.062	0.371
M5-aug	0.300	0.307	0.348	0.268	0.892	0.933	0.661	0.095	0.079	0.483
m6r16	0.297	0.303	0.343	0.274	0.689	0.807	0.435	0.092	0.118	0.478
m6r16-aug	0.298	0.305	0.346	0.270	0.854	0.916	0.632	0.101	0.105	0.551
m6r64	0.296	0.301	0.345	0.275	0.719	0.826	0.371	0.121	0.096	0.439
m6r64-aug	0.297	0.304	0.346	0.272	0.881	0.933	0.619	0.101	0.106	0.510
m6r256	0.296	0.302	0.344	0.273	0.750	0.846	0.428	0.116	0.103	0.460
m6r256-aug	0.297	0.303	0.344	0.271	0.888	0.942	0.622	0.099	0.105	0.508
m6	0.303	0.308	0.346	0.258	0.726	0.830	0.382	0.132	0.114	0.429
m6-aug	0.304	0.309	0.348	0.257	0.863	0.930	0.606	0.108	0.104	0.504
m6r	0.298	0.304	0.346	0.270	0.786	0.868	0.474	0.130	0.095	0.457
m6r-aug	0.300	0.306	0.346	0.266	0.893	0.954	0.661	0.153	0.094	0.476
Spread over seeds										
M3k	0.003	0.001	0.001	0.003	0.012	0.004	0.001	0.003	0.006	0.015
M3k, 256	0.001	0.001	0.001	0.000	0.067	0.055	0.034	0.001	0.008	0.023
M5	0.003	0.002	0.002	0.002	0.021	0.022	0.030	0.038	0.042	0.097
M5-aug	0.003	0.000	0.003	0.002	0.024	0.039	0.070	0.018	0.014	0.091
m6r16	0.002	0.003	0.001	0.001	0.016	0.022	0.023	0.006	0.024	0.038
m6r16-aug	0.003	0.002	0.003	0.003	0.065	0.041	0.095	0.024	0.024	0.076
m6r64	0.001	0.001	0.002	0.004	0.030	0.019	0.069	0.034	0.001	0.032
m6r64-aug	0.002	0.001	0.002	0.001	0.049	0.020	0.112	0.015	0.014	0.020
m6r256	0.001	0.003	0.001	0.001	0.011	0.008	0.020	0.021	0.013	0.045
m6r256-aug	0.003	0.001	0.000	0.001	0.072	0.029	0.149	0.030	0.037	0.019
m6	0.004	0.002	0.003	0.000	0.034	0.018	0.061	0.019	0.036	0.008
m6-aug	0.003	0.002	0.002	0.004	0.059	0.026	0.111	0.013	0.094	0.017
m6r	0.001	0.002	0.002	0.003	0.021	0.006	0.037	0.024	0.005	0.054
m6r-aug	0.001	0.001	0.002	0.005	0.034	0.014	0.059	0.072	0.029	0.066

Table 8: The far stack and the nearest-neighbour filters against the exclusion radius R, validation years: mask error on `scatter`, noise removed, median retention of 25- and 100-km transients and median largest spurious change.

	R (km)							
Filter	0	10	25	50	100	200	400	600
Mask error								
Far stack	0.302	0.302	0.303	0.303	0.304	0.307	0.316	0.375
M1	0.320	0.319	0.320	0.328	0.339	0.350	0.346	–
M1, 64	0.302	0.304	0.305	0.309	0.316	0.324	0.319	–
M1, 256	0.297	0.299	0.299	0.300	0.301	0.307	0.317	–
M2	0.319	0.321	0.321	0.327	0.337	0.337	0.344	–
M3a	0.323	0.324	0.324	0.329	0.336	0.342	0.344	–
M3b	0.318	0.316	0.317	0.319	0.330	0.337	0.335	–
M4a	0.318	0.321	0.323	0.331	0.328	0.327	0.331	–
M4b	0.314	0.316	0.316	0.319	0.329	0.331	0.330	–
Noise removed								
Far stack	0.268	0.267	0.267	0.266	0.264	0.260	0.242	0.148
M1	0.251	0.249	0.247	0.238	0.217	0.208	0.204	–
M1, 64	0.268	0.264	0.262	0.252	0.247	0.238	0.236	–
M1, 256	0.276	0.274	0.273	0.271	0.269	0.260	0.241	–
M2	0.231	0.229	0.229	0.220	0.204	0.198	0.185	–
M3a	0.240	0.232	0.230	0.218	0.208	0.201	0.188	–
M3b	0.239	0.235	0.233	0.222	0.212	0.203	0.194	–
M4a	0.256	0.248	0.245	0.232	0.227	0.225	0.213	–
M4b	0.245	0.239	0.235	0.227	0.215	0.214	0.205	–
Kept, $L = 25$ km								
Far stack	0.985	0.986	0.990	0.997	1.000	1.000	1.000	1.000
M1	0.471	0.613	0.741	0.928	0.999	1.000	1.000	–
M1, 64	0.816	0.832	0.892	0.967	1.000	1.000	1.000	–
M1, 256	0.878	0.948	0.964	0.976	1.000	1.000	1.000	–
M2	0.585	0.635	0.789	0.935	0.999	1.000	1.000	–
M3a	0.635	0.687	0.795	0.946	0.999	1.000	1.000	–
M3b	0.602	0.638	0.767	0.932	0.999	1.000	1.000	–
M4a	0.652	0.692	0.807	0.942	1.000	1.000	1.000	–
M4b	0.608	0.636	0.766	0.933	0.999	1.000	1.000	–
Kept, $L = 100$ km								
Far stack	0.774	0.776	0.779	0.793	0.832	0.936	0.999	1.000
M1	0.025	0.039	0.049	0.084	0.240	0.681	0.996	–
M1, 64	0.132	0.136	0.146	0.183	0.346	0.765	0.998	–
M1, 256	0.260	0.446	0.463	0.315	0.633	0.864	0.998	–
M2	0.030	0.036	0.053	0.083	0.258	0.691	0.996	–
M3a	0.204	0.206	0.218	0.248	0.381	0.750	0.997	–
M3b	0.086	0.102	0.104	0.127	0.290	0.719	0.996	–
M4a	0.205	0.212	0.234	0.261	0.478	0.804	0.997	–
M4b	0.110	0.106	0.102	0.140	0.305	0.755	0.997	–
Largest spurious change								
Far stack	0.028	0.029	0.029	0.030	0.035	0.047	0.095	0.123
M1	0.088	0.113	0.119	0.144	0.229	0.316	0.383	–
M1, 64	0.113	0.113	0.116	0.123	0.161	0.185	0.181	–
M1, 256	0.110	0.103	0.102	0.123	0.106	0.105	0.160	–
M2	0.218	0.226	0.241	0.269	0.414	0.566	0.699	–
M3a	0.105	0.109	0.115	0.143	0.232	0.307	0.375	–
M3b	0.120	0.128	0.135	0.168	0.243	0.332	0.418	–
M4a	0.109	0.111	0.116	0.146	0.198	0.257	0.309	–
M4b	0.119	0.125	0.132	0.165	0.243	0.298	0.363	–

Table 9: Learned filters against the exclusion radius, seed 0, validation years: mask error on `scatter` at every trained radius, and retention over all transients, retention of 100-km transients and largest spurious change at $R = 0$ and 25 km.

	Mask error, R (km)							Kept		Kept, 100 km		Spurious	
Filter	0	10	25	50	100	200	400	0	25	0	25	0	25
M3k	0.314	0.315	0.313	0.315	0.325	0.331	0.333	0.30	0.45	0.04	0.05	0.10	0.12
M3k, 256	0.293	0.296	0.294	0.296	0.299	0.306	0.315	0.71	0.72	0.32	0.28	0.11	0.12
M5	0.298	0.296	0.299	0.300	0.303	0.307	0.315	0.71	0.90	0.41	0.60	0.13	0.10
M5-aug	0.299	0.300	0.302	0.300	0.305	0.307	0.314	0.89	0.95	0.66	0.71	0.08	0.14
m6r16	0.297	0.299	0.300	0.302	0.302	0.308	0.315	0.70	0.77	0.45	0.47	0.09	0.10
m6r16-aug	0.300	0.303	0.300	0.301	0.304	0.308	0.319	0.88	0.91	0.68	0.64	0.12	0.14
m6r64	0.296	0.298	0.299	0.299	0.303	0.305	0.314	0.71	0.82	0.37	0.44	0.12	0.12
m6r64-aug	0.296	0.299	0.300	0.299	0.304	0.305	0.314	0.88	0.91	0.62	0.57	0.09	0.10
m6r256	0.296	0.297	0.299	0.300	0.302	0.306	0.313	0.75	0.83	0.42	0.47	0.12	0.11
m6r256-aug	0.296	0.301	0.298	0.303	0.304	0.306	0.314	0.93	0.94	0.70	0.70	0.12	0.10
m6	0.301	0.303	0.305	0.306	0.309	–	–	0.71	0.82	0.35	0.43	0.14	0.14
m6-aug	0.302	0.303	0.306	0.306	0.310	–	–	0.86	0.91	0.60	0.64	0.12	0.11
m6r	0.298	0.297	0.301	0.300	0.303	–	–	0.79	0.87	0.47	0.53	0.14	0.12
m6r-aug	0.301	0.300	0.303	0.302	0.304	–	–	0.90	0.91	0.67	0.62	0.12	0.11

regressions keep the most (M3a 0.210, M3b 0.191, M4a 0.187, M4b 0.189); M1 and M2 keep 0.095 and 0.083. The far stack keeps 0.125–0.137 from $R = 0$ to 200 km, 0.169 at 400 km and 0.355 at 600 km. At 400 km, where it keeps 0.999 of 100-km planted transients, it keeps 0.097 of the horizontal and 0.170 of the vertical seasonal cycle. For the learned filters, the seed spread of seasonal kept (up to 0.094) is comparable to the range of their seed means (0.062–0.132).

6.6. Real earthquake offset

The validation years contain one offset that passes the selection: 4.89 mm at IDNR from the 31 March 2020 M6.5 central Idaho earthquake. The unfiltered series gives 1.00 by construction. Every filter except two C3 settings reports more than the observed offset: the far stack 1.461 (1.461–1.509 over its radii), C1 1.457, the filters over 16 neighbours 1.26–1.32 and the learned filters 1.30–1.52 (seed means 1.31–1.48). C3 with 4 regions and 3 components keeps 0.812, and with 8 regions and 3 components 0.404. With one offset at one station, this measure has no statistical weight on the validation years.

6.7. Slow-slip episodes

The validation run scores 18 episodes of the GNSS catalogue of Michel et al. (2019) with the revised measure of Section 4.2 (Table 10). Fifteen have

Table 10: Slow-slip episodes scored on the validation run (GNSS catalogue of Michel et al., 2019, training years, excluded from training), revised measure: number of scored band stations, their median slow-slip estimate, the noise floor, and the pooled fraction kept for selected filters, $R = 0$, seed 0. Episodes with fewer than 10 stations (*) and episodes below twice the floor (†) are left out of the summary.

Episode	Sta.	Est. (mm)	Floor (mm)	Far stack	C1	M1	M1, 256	M3a	M3k, 256	M5-aug	m6r256-aug
M19-14_15†	159	0.93	0.55	0.98	0.97	0.87	0.95	0.77	0.94	0.85	0.91
M19-16_18_19	132	3.35	0.88	0.70	0.70	0.33	0.44	0.22	0.46	0.47	0.41
M19-22+23*	2	1.77	0.50	0.98	0.97	0.55	0.66	0.73	0.60	0.62	0.78
M19-24	58	4.00	0.68	0.84	0.83	0.19	0.41	0.28	0.40	0.45	0.34
M19-26*	7	1.96	0.70	1.03	1.02	0.41	0.70	0.47	0.72	0.70	0.76
M19-27_28_29_30	138	3.20	0.92	0.76	0.76	0.40	0.51	0.25	0.52	0.53	0.48
M19-34_36_37†	196	2.12	1.18	1.00	1.00	0.92	0.97	0.83	0.96	0.92	0.93
M19-38+39	69	3.91	0.79	0.84	0.83	0.34	0.51	0.34	0.53	0.38	0.34
M19-40_41	66	2.67	0.76	1.03	1.03	0.54	0.74	0.30	0.74	0.60	0.62
M19-43_44†	101	1.29	0.83	0.89	0.89	0.47	0.63	0.51	0.62	0.63	0.67
M19-45+46	47	2.83	1.02	0.88	0.88	0.42	0.68	0.24	0.65	0.58	0.55
M19-47+50_48†	137	1.29	0.78	0.94	0.94	0.69	0.86	0.44	0.83	0.81	0.80
M19-51	51	2.56	0.74	0.87	0.87	0.35	0.62	0.30	0.60	0.56	0.61
M19-53*	2	1.00	0.91	1.33	1.32	0.39	1.13	0.71	1.08	1.15	0.80
M19-56_54+55	161	4.43	0.89	0.68	0.68	0.36	0.47	0.45	0.47	0.53	0.60
M19-59	63	3.48	1.35	0.80	0.80	0.19	0.44	0.25	0.42	0.39	0.46
M19-62+63†	176	0.99	0.71	0.99	0.98	0.88	0.96	0.55	0.95	0.89	0.89
M19-9_10	147	2.63	1.08	0.88	0.88	0.65	0.75	0.44	0.76	0.70	0.68
Summary (10 episodes)	932			0.836	0.833	0.357	0.509	0.287	0.525	0.530	0.513

at least 10 scored stations. Ten of these are counted (932 station-episodes); five fall below twice their noise floor and are left out (M19-14_15, M19-34_36_37, M19-43_44, M19-47+50_48, M19-62+63). The median slow-slip estimate of the counted episodes is 2.6–4.4 mm, against noise floors of 0.50–1.35 mm over all 18.

The far stack keeps 0.836 (median over the counted episodes), C1 0.833, D1 at 400 km 1.006 and the far stack at 600 km 1.012. The unfiltered series scores 0.888, not 1, because it still holds the reference motion. At $R = 0$ the filters over nearest neighbours keep 0.25–0.51 (M4a the least, M1 over 256 neighbours the most), and at 400 km 0.78–1.00. The sub-regional pPCA keeps 0.79 with one region and one component and 0.33 with eight regions and three components. The learned filters keep 0.36–0.55 (seed means; M3k the least, m6r16-aug the most), and their seed spread reaches 0.097. Per counted episode, the far stack's value ranges from 0.68 (M19-56_54+55) to 1.03 (M19-40_41).

6.8. Training record

Table 11 gives, for every learned filter at $R = 0$, the step (epoch for M3k) at which the validation criterion was best and its value, for each seed. The

Table 11: Training of the learned filters at $R = 0$: best step (attention networks) or epoch (M3k) and the validation criterion there, seeds 0, 1 and 2.

	Best step or epoch			Validation criterion		
Filter	0	1	2	0	1	2
M3k	2	1	1	0.3246	0.3242	0.3245
M3k, 256	1	1	1	0.3113	0.3117	0.3119
M5	8,500	3,500	6,000	0.3171	0.3173	0.3168
M5-aug	3,000	6,000	4,500	0.3198	0.3188	0.3195
m6r16	3,000	3,500	4,500	0.3169	0.3166	0.3165
m6r16-aug	7,000	3,500	4,000	0.3181	0.3183	0.3175
m6r64	4,000	7,500	5,500	0.3158	0.3150	0.3155
m6r64-aug	5,500	2,500	8,000	0.3167	0.3172	0.3167
m6r256	7,500	9,000	6,500	0.3159	0.3161	0.3165
m6r256-aug	9,000	4,500	3,500	0.3163	0.3165	0.3173
m6	3,000	2,500	3,000	0.3185	0.3193	0.3193
m6-aug	3,000	2,000	3,500	0.3196	0.3188	0.3196
m6r	2,500	2,000	3,500	0.3184	0.3177	0.3177
m6r-aug	5,500	5,500	8,500	0.3176	0.3183	0.3175

validation criterion is computed on every available validation day of the seen, scored stations and is higher than the `scatter` mask error of the same model (0.311–0.325 against 0.292–0.314). M3k reached its best criterion at the first or second epoch at $R = 0$ for every seed, and over 256 neighbours at the first epoch for every seed; at other radii its best epochs were 1–7 over 16 neighbours and 1–17 over 256. The attention networks reached their best at 2,000–9,000 steps at $R = 0$. Every one of the 118 training runs ended without error and passed both leak checks.

6.9. Success criterion

Table 12 applies the criterion of Section 4.4 at $R = 0$. The reference, the far stack at $R = 0$, removes 0.268 of the short-period scatter, keeps 0.774 of 100-km planted transients and has a spurious signal of 0.157 for them. Of the 132 other scored combinations of filter, seed and radius, one is better, 19 are noise gains bought with signal and 112 are not better. The one that passes is C1, the network mean with the target included: its noise removed is higher by 0.002, with an interval of no width because the two filters differ only in the target station's own share of the mean. The noise gains bought with signal are, by filter and number of combinations: M3k, 256 (4), m6r16 (3), M1, 256 (2), m6r64 (2), c3-k4p2 (1), c3-k4p3 (1), c3-k8p2 (1), c3-k8p3

(1), M2-self (1), M5 (1), m6r256 (1), m6r64-aug (1). No learned filter passes at any seed; the planted-transient results used here predate the change of identifiers (Section 4.4), which does not affect this table because it pairs nothing by identifier.

6.10. Neighbours read

Table 13 gives, for each exclusion radius and neighbour count N, the number of the 464 scored stations with fewer than N neighbours, the median number read and the median distance to the farthest neighbour read. No station has fewer than four neighbours at any radius, so none is left out of scoring. With $N = 16$ every station has its 16 neighbours at every radius. With $N = 256$ more than half the stations fall short at 0, 10, 200 and 400 km; at 400 km the median filter reads 132 stations.

7. Open questions

These are questions the validation numbers raise, listed for the author's decision; none has been acted on.

1. Seasonal kept is low for every filter, including the far stack at 400 km, which keeps every planted transient (Section 6.5). A check on 12 November 2026 on the validation years showed that this is the network-wide part of the cycle: the annual amplitude of the plain network mean is 2.2, 3.6 and 5.8 mm (east, north, up) against station medians of 2.40, 3.75 and 6.14 mm, and subtracting the harmonic of the network mean from each station's leaves a median of 0.128 of it, against 0.125 kept by the far stack. Whether a measure of the station-specific part of the cycle is wanted is open. The low values of the first episode measure had the same origin and led to the revised measure (Section 4.2).
2. The only filter that passes the success criterion, C1, differs from the reference only in including the target station; its gain in noise removed is 0.002 (Section 6.9). The criterion sets no smallest gain.
3. The real-offset measure rests on one offset at one station on the validation years, and 33 of 35 filters report it as larger than observed (Section 6.6). Whether to keep it, widen its selection, or report it only on the test years is open.
4. The slow-slip exclusion bars 13.5% of the available training station-days (Table 2), and there is no catalogued GNSS episode list for 2017–2022

Table 12: Success criterion at $R = 0$ (D1: 400 km), validation years, seed 0 for the learned filters. Noise: noise removed. Difference: from the far stack at $R = 0$, with its 95% bootstrap interval over stations and 30-day blocks. Kept and spurious: 100-km planted transients. A filter is better if the interval lies above zero, kept is at least 0.724 and spurious at most 0.207.

Filter	Noise	Difference [95% interval]	Kept	Spurious	Verdict
Target-inclusive					
C1	0.270	+0.002 [+0.002, +0.002]	0.772	0.156	better
c3-k1p1	0.259	−0.009 [−0.015, −0.001]	0.763	0.268	not better
c3-k1p2	0.263	−0.005 [−0.014, +0.001]	0.756	0.289	not better
c3-k1p3	0.263	−0.005 [−0.014, +0.001]	0.732	0.379	not better
c3-k4p1	0.276	+0.008 [−0.003, +0.014]	0.476	0.331	not better
c3-k4p2	0.278	+0.010 [+0.001, +0.017]	0.453	0.331	bought with signal
c3-k4p3	0.282	+0.014 [+0.004, +0.022]	0.427	0.333	bought with signal
c3-k8p1	0.279	+0.011 [0.000, +0.019]	0.220	0.153	not better
c3-k8p2	0.289	+0.021 [+0.010, +0.029]	0.216	0.155	bought with signal
c3-k8p3	0.295	+0.027 [+0.016, +0.038]	0.206	0.158	bought with signal
M2-self	0.291	+0.023 [+0.015, +0.035]	0.026	0.098	bought with signal
Distance-excluded					
Far stack	0.268	0.000 [0.000, 0.000]	0.774	0.157	reference
D1 (400 km)	0.242	−0.026 [−0.031, −0.020]	0.999	0.344	not better
Leave-one-out over nearest neighbours					
M1	0.251	−0.017 [−0.029, −0.008]	0.025	0.027	not better
M1, 64	0.268	0.000 [−0.009, +0.010]	0.132	0.083	not better
M1, 256	0.276	+0.008 [+0.001, +0.016]	0.260	0.111	bought with signal
M2	0.231	−0.037 [−0.046, −0.026]	0.030	0.103	not better
M3a	0.240	−0.028 [−0.042, −0.021]	0.204	0.038	not better
M3b	0.239	−0.029 [−0.040, −0.021]	0.086	0.042	not better
M4a	0.256	−0.012 [−0.024, −0.005]	0.205	0.040	not better
M4b	0.245	−0.023 [−0.035, −0.016]	0.110	0.044	not better
Learned, seed 0					
M3k	0.258	−0.010 [−0.020, −0.001]	0.035	0.039	not better
M3k, 256	0.284	+0.016 [+0.012, +0.024]	0.323	0.175	bought with signal
M5	0.270	+0.002 [−0.005, +0.008]	0.411	0.192	not better
M5-aug	0.269	+0.001 [−0.004, +0.007]	0.663	0.136	not better
m6r16	0.274	+0.006 [+0.001, +0.012]	0.448	0.134	bought with signal
m6r16-aug	0.268	0.000 [−0.004, +0.007]	0.677	0.210	not better
m6r64	0.276	+0.008 [+0.002, +0.013]	0.373	0.178	bought with signal
m6r64-aug	0.273	+0.005 [0.000, +0.011]	0.622	0.149	bought with signal
m6r256	0.272	+0.004 [0.000, +0.012]	0.423	0.220	bought with signal
m6r256-aug	0.271	+0.003 [−0.001, +0.010]	0.699	0.175	not better
m6	0.258	−0.010 [−0.018, −0.001]	0.354	0.203	not better
m6-aug	0.255	−0.013 [−0.020, −0.003]	0.604	0.165	not better
m6r	0.270	+0.002 [−0.006, +0.006]	0.471	0.253	not better
m6r-aug	0.268	0.000 [−0.006, +0.007]	0.669	0.244	not better

Table 13: Neighbours actually read on the Pacific Northwest cube, 464 scored stations: stations with fewer than N neighbours beyond R (short), median number read, and median distance to the farthest neighbour read (reach, km). *: more than half the stations are short.

	$N = 16$			$N = 64$			$N = 256$		
R (km)	Short	Read	Reach	Short	Read	Reach	Short	Read	Reach
0	0	16	67	17	64	136	261	244	297*
10	0	16	69	15	64	138	236	253	306*
25	0	16	72	12	64	140	219	256	314
50	0	16	84	10	64	147	192	256	324
100	0	16	122	2	64	175	166	256	361
200	0	16	218	0	64	265	243	254	495*
400	0	16	420	52	64	491	405	132	676*

to score or exclude. Five of the 15 scored episodes fall below the floor rule of the revised measure.

5. The architectures and training settings of the learned filters were carried over from California. M3k stopped at its first or second epoch at $R = 0$ on every seed (Section 6.8).
6. The scoring QC reads each station's own series in the test years, and six of the eight stations it leaves out were flagged by a jump first seen there (Section 3.2).
7. The planted-transient results predate the change of the transient identifiers (Section 4.4); they are being produced again, together with training seeds 1 and 2 and the signal tests at every radius.

Data and code availability

The daily solutions are publicly available from the Nevada Geodetic Laboratory (Nevada Geodetic Laboratory, 2026). The `gnssdl` package is openly available at `https://github.com/hogib/gnssdl` under the MIT licence; the numbers of this report are reproduced from the results files in `data/runs/` by the script `summary.py` in `notes/internal/work/`.

References

Bachelot, L., Thomas, A., Melgar, D., Searcy, J., Sun, Y.S., 2025. Cascadia Daily GNSS Time Series Denoising: Graph Neural Network and Stack Filtering. Seismica 4. doi:10.26443/seismica.v2i4.1419.

Batson, J., Royer, L., 2019. Noise2Self: Blind Denoising by Self-Supervision, in: Proceedings of the 36th International Conference on Machine Learning, pp. 524–533.

Bedford, J., Bevis, M., 2018. Greedy Automatic Signal Decomposition and Its Application to Daily GPS Time Series. Journal of Geophysical Research: Solid Earth 123, 6992–7003. doi:10.1029/2017JB014765.

Bletery, Q., Nocquet, J.M., 2023. The precursory phase of large earthquakes. Science 381, 297–301. doi:10.1126/science.adg2565.

Bletery, Q., Nocquet, J.M., 2025. Do large earthquakes start with a precursory phase of slow slip? Seismica 3. doi:10.26443/seismica.v3i2.1383.

Blewitt, G., Hammond, W., Kreemer, C., 2018. Harnessing the GPS Data Explosion for Interdisciplinary Science. Eos 99. doi:10.1029/2018EO104623.

Cini, A., Marisca, I., Alippi, C., 2022. Filling the G_ap_s: Multivariate Time Series Imputation by Graph Neural Networks, in: International Conference on Learning Representations (ICLR). ArXiv:2108.00298.

Costantino, G., Giffard-Roisin, S., Dalla Mura, M., Socquet, A., 2024. Denoising of Geodetic Time Series Using Spatiotemporal Graph Neural Networks: Application to Slow Slip Event Extraction. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 17, 17567–17579. doi:10.1109/JSTARS.2024.3465270.

Costantino, G., Giffard-Roisin, S., Radiguet, M., Dalla Mura, M., Marsan, D., Socquet, A., 2023. Multi-station deep learning on geodetic time series detects slow slip events in Cascadia. Communications Earth & Environment 4, 435. doi:10.1038/s43247-023-01107-7.

Costantino, G., Radiguet, M., El Yousfi, Z., Socquet, A., 2026. A Continuum of Slow Slip Events in the Cascadia Subduction Zone Illuminated by High-Resolution Deep-Learning Denoising. Geophysical Research Letters 53, e2025GL117446. doi:10.1029/2025GL117446.

Crawford, W.C., Webb, S.C., 2000. Identifying and Removing Tilt Noise from Low-Frequency (<0.1 Hz) Seafloor Vertical Seismic Data. Bulletin of the Seismological Society of America 90, 952–963. doi:10.1785/0119990121.

Dong, D., Fang, P., Bock, Y., Webb, F., Prawirodirdjo, L., Kedar, S., Jamason, P., 2006. Spatiotemporal filtering using principal component analysis and Karhunen-Loeve expansion approaches for regional GPS network analysis. Journal of Geophysical Research: Solid Earth 111, 2005JB003806. doi:10.1029/2005JB003806.

van den Ende, M., Lior, I., Ampuero, J.P., Sladen, A., Ferrari, A., Richard, C., 2023. A Self-Supervised Deep Learning Approach for Blind Denoising and Waveform Coherence Enhancement in Distributed Acoustic Sensing Data. IEEE Transactions on Neural Networks and Learning Systems 34, 3371–3384. doi:10.1109/tnnls.2021.3132832.

Gonzalez, T.F., 1985. Clustering to minimize the maximum intercluster distance. Theoretical Computer Science 38, 293–306. doi:10.1016/0304-3975(85)90224-5.

Gruszczynski, M., Klos, A., Bogusz, J., 2018. A Filtering of Incomplete GNSS Position Time Series with Probabilistic Principal Component Analysis. Pure and Applied Geophysics 175, 1841–1867. doi:10.1007/s00024-018-1856-3.

Gualandi, A., Serpelloni, E., Belardinelli, M.E., 2016. Blind source separation problem in GPS time series. Journal of Geodesy 90, 323–341. doi:10.1007/s00190-015-0875-4.

Janiszewski, H.A., Gaherty, J.B., Abers, G.A., Gao, H., Eilon, Z.C., 2019. Amphibious surface-wave phase-velocity measurements of the Cascadia subduction zone. Geophysical Journal International 217, 1929–1948. doi:10.1093/gji/ggz051.

Ji, K.H., Herring, T.A., 2013. A method for detecting transient signals in GPS position time-series: smoothing and principal component analysis. Geophysical Journal International 193, 171–186. doi:10.1093/gji/ggt003.

Kreemer, C., Blewitt, G., 2021. Robust estimation of spatially varying common-mode components in GPS time-series. Journal of Geodesy 95, 13. doi:10.1007/s00190-020-01466-5.

Le Rest, K., Pinaud, D., Monestiez, P., Chadoeuf, J., Bretagnolle, V., 2014. Spatial leave-one-out cross-validation for variable selection in the presence

of spatial autocorrelation. Global Ecology and Biogeography 23, 811–820. doi:10.1111/geb.12161.

Li, W., Jiang, W., Li, Z., Chen, H., Chen, Q., Wang, J., Zhu, G., 2020. Extracting Common Mode Errors of Regional GNSS Position Time Series in the Presence of Missing Data by Variational Bayesian Principal Component Analysis. Sensors 20, 2298. doi:10.3390/s20082298.

Liu, B., Jiang, Y., Xiao, Z., Rao, W., Feng, T., 2026. An adaptive spatiotemporal filtering method for GNSS coordinate time series of CMONOC. Earth, Planets and Space 78, 158. doi:10.1186/s40623-026-02477-w.

Marisca, I., Cini, A., Alippi, C., 2022. Learning to Reconstruct Missing Data from Spatiotemporal Graphs with Sparse Observations, in: Advances in Neural Information Processing Systems (NeurIPS). ArXiv:2205.13479.

Mastella, G., Bedford, J., Corbi, F., Funiciello, F., 2025. Denoising daily displacement GNSS time series using deep neural networks in a near real-time framing: a single-station method. Geophysical Journal International 242, ggaf207. doi:10.1093/gji/ggaf207.

Michel, S., Gualandi, A., Avouac, J.P., 2019. Similar scaling laws for earthquakes and Cascadia slow-slip events. Nature 574, 522–526. doi:10.1038/s41586-019-1673-6.

Nevada Geodetic Laboratory, 2026. GPS Networks Map Viewer and Time Series Products (IGS20 daily tenv3). https://geodesy.unr.edu. Accessed 6 October 2026.

Ohtani, R., McGuire, J.J., Segall, P., 2010. Network strain filter: A new tool for monitoring and detecting transient deformation signals in GPS arrays. Journal of Geophysical Research: Solid Earth 115. doi:10.1029/2010JB007442.

Okada, Y., 1992. Internal deformation due to shear and tensile faults in a half-space. Bulletin of the Seismological Society of America 82, 1018–1040. doi:10.1785/BSSA0820021018.

Riel, B., Simons, M., Agram, P., Zhan, Z., 2014. Detecting transient signals in geodetic time series using sparse estimation techniques. Journal of Geophysical Research: Solid Earth 119, 5140–5160. doi:10.1002/2014JB011077.

Rouet-Leduc, B., Jolivet, R., Dalaison, M., Johnson, P.A., Hulbert, C., 2021. Autonomous extraction of millimeter-scale deformation in InSAR time series using deep learning. Nature Communications 12, 6480. doi:10.1038/s41467-021-26254-3.

Shen, Y., Li, W., Xu, G., Li, B., 2014. Spatiotemporal filtering of regional GNSS network's position time series with missing data using principle component analysis. Journal of Geodesy 88, 1–12. doi:10.1007/s00190-013-0663-y.

Tanaka, Y., Kano, M., Yano, K., 2025. Detecting Slow Slip Signals in Southwest Japan Based on Machine Learning Trained by Real GNSS Time Series. Journal of Geophysical Research: Solid Earth 130, e2024JB029499. doi:10.1029/2024JB029499.

Teutschmann, N., Crocetti, L., Lehmann, F., Trentini, L., Soja, B., 2026. GNSS-DispFM: A Self-Supervised Foundation Model for Daily GNSS Displacement Time Series. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing doi:10.1109/JSTARS.2026.3740122. early access.

Tian, Y., Shen, Z.K., 2016. Extracting the regional common-mode component of GPS station position time series from dense continuous network. Journal of Geophysical Research: Solid Earth 121, 1080–1096. doi:10.1002/2015JB012253.

Tipping, M.E., Bishop, C.M., 1999. Probabilistic Principal Component Analysis. Journal of the Royal Statistical Society Series B: Statistical Methodology 61, 611–622. doi:10.1111/1467-9868.00196.

Wdowinski, S., Bock, Y., Zhang, J., Fang, P., Genrich, J., 1997. Southern California permanent GPS geodetic array: Spatial filtering of daily positions for estimating coseismic and postseismic displacements induced by the 1992 Landers earthquake. Journal of Geophysical Research: Solid Earth 102, 18057–18070. doi:10.1029/97JB01378.

Wu, Y., Zhuang, D., Labbe, A., Sun, L., 2021. Inductive Graph Neural Networks for Spatiotemporal Kriging, in: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 4478–4485. doi:10.1609/aaai.v35i5.16575.

Zhou, W., Ding, K., Liu, P., Lan, G., Ming, Z., 2022. Spatiotemporal Filtering for Continuous GPS Coordinate Time Series in Mainland China by Using Independent Component Analysis. Remote Sensing 14, 2904. doi:10.3390/rs14122904.